Supplementary Information

for

Delay selection by spike-timing-dependent plasticity shapes efficient networks for signal transmission

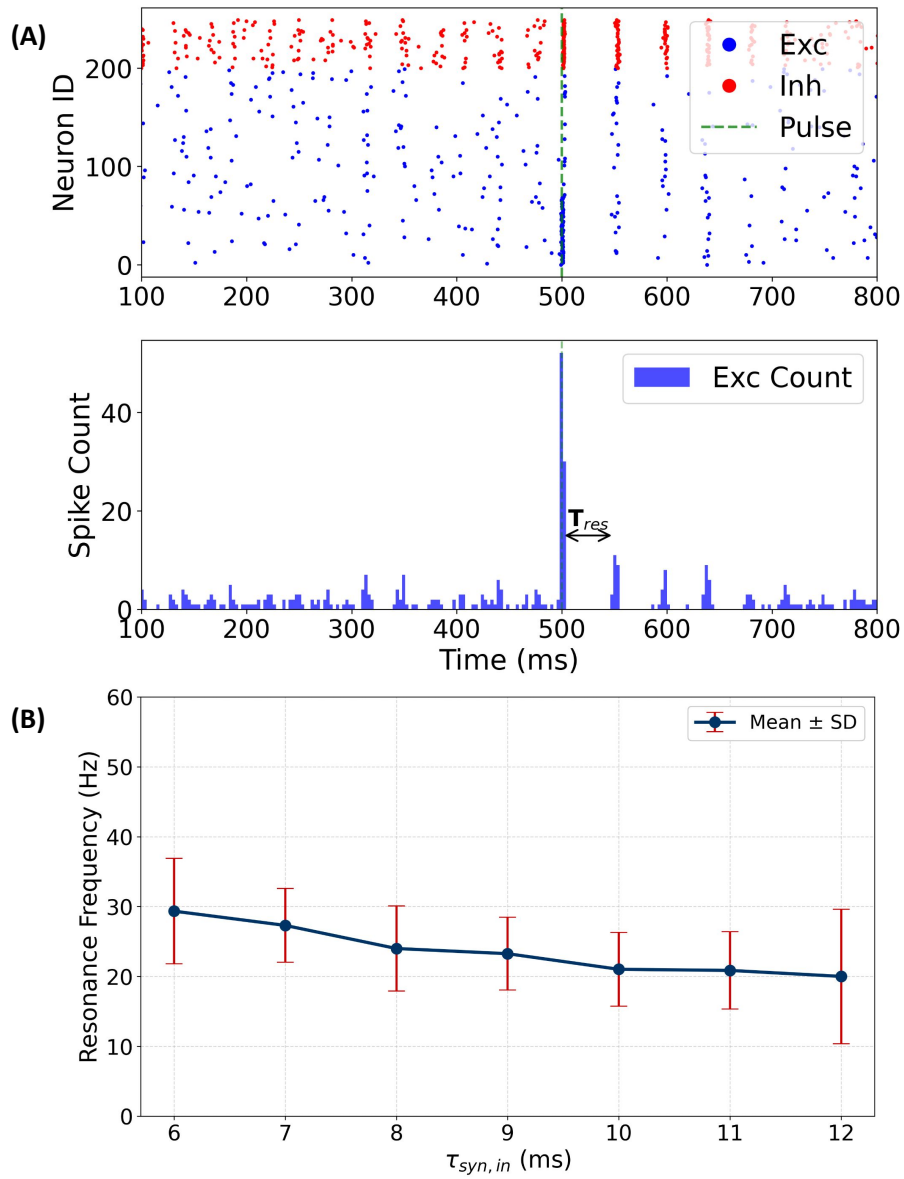


Figure S1: Quantification of pulse-induced resonance frequency and its dependence on inhibitory synaptic kinetics. (A) Characterization of the response of a single network to a transient stimulus. The top panel shows a representative spike raster of excitatory (blue) and inhibitory (red) populations; the bottom panel displays the corresponding population activity (bin size = 3 ms). Prior to the stimulus (t < 500 ms), the network resides in an inhibition-dominated regime, characterized by small amplitude fluctuations of network activity, weak synchrony and oscillations (which can be more prominently seen in the inhibitory population). The arrival of a pulse packet at t = 500 ms (green dashed line) triggers a transient, high-amplitude oscillation. The network resonance frequency (f_{res}) is defined as the reciprocal of the period (T) of this emergent oscillation. **(B)** Sensitivity of the network resonance frequency to local inhibitory kinetics. The resonance frequency is plotted as a function of the inhibitory synaptic time constant (τ_{in}). Blue markers and red error bars indicate the mean and the standard deviation, calculated for 101 individual realizations (n = 101 random seeds per parameter point). The results demonstrate that increasing the inhibitory decay time constant systematically decreases the network resonance frequency, demonstrating that the network's resonance frequency is directly tunable via the inhibitory synaptic decay time constant.

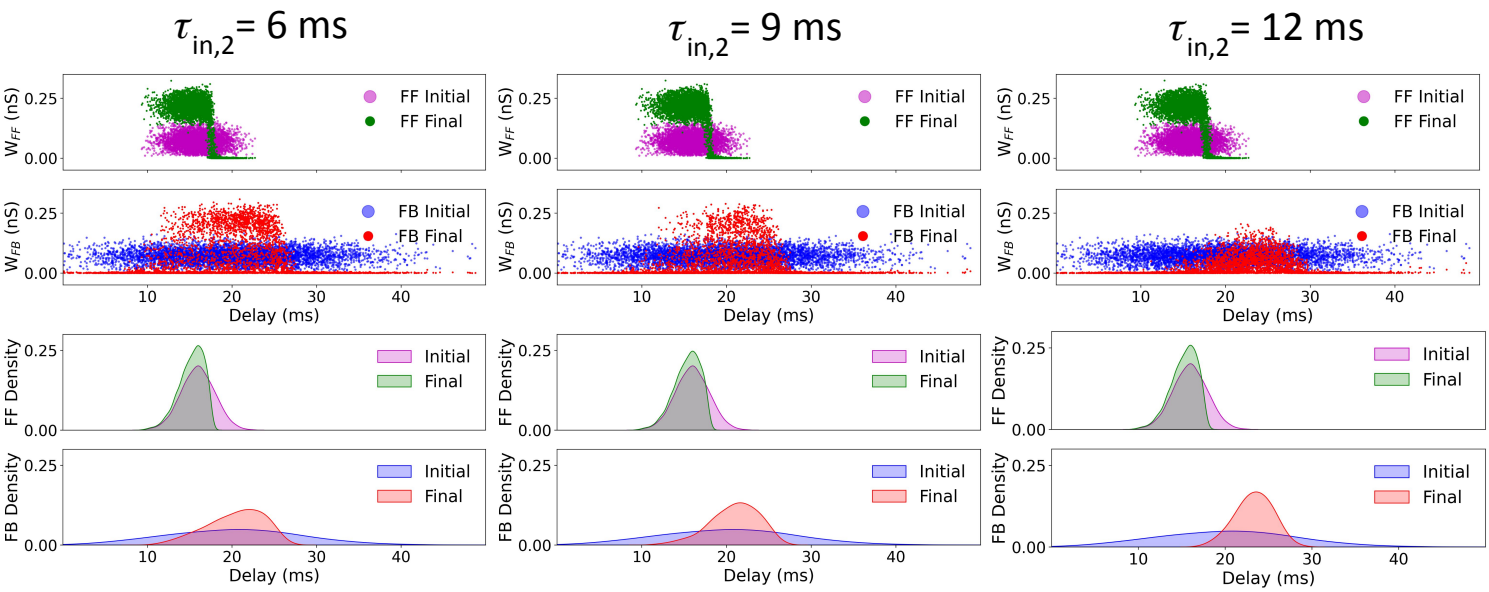


Figure S2: Selection of axonal delays governed by inter-layer resonance frequency mismatch. (Upper panels): Scatter plots comparing initial and final states for feedforward (FF) and feedback (FB) connections. **(Lower panels)** Final delay distributions for FF and FB connections across varying inhibitory synaptic time constants in Layer 2 ($\tau_{in} \in$ [6, 12] ms), where τ_{in} for the layer 1 is fixed at 8 ms. The FF pathway exhibits a stable delay cutoff (approx. 22–24 ms) across all conditions, indicating a robust temporal window for feedforward potentiation that is largely independent of Layer 2 dynamics. Conversely, the FB connections demonstrate high sensitivity to the network's resonance state: the FB delay cutoff is less abrupt and systematically shifts toward higher values as the resonance period of layer 2 increases. Notably, at the highest mismatch (τ_{in} = 12 ms that leads to a resonance frequency ~20 Hz for the layer 2 network), potentiation in the FB pathway is nearly abolished, with very few connections reaching the strength threshold to be considered in the final delay distribution. These results suggest that while the feedforward connectivity is governed by a fixed temporal filter, the feedback loop undergoes a dynamic reorganization that tracks the emergent oscillation period, eventually failing to stabilize when inter-layer resonance is sufficiently misaligned.

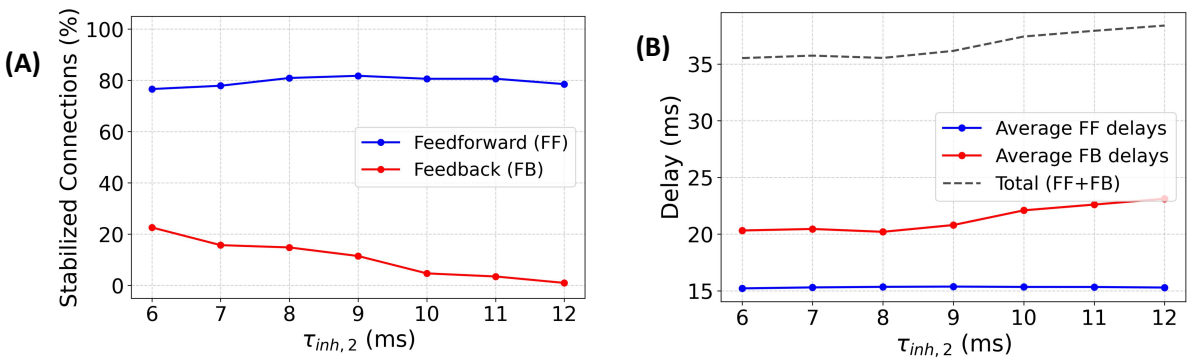


Figure S3: Impact of resonance frequency mismatch on delay selection. (A) Percentage of potentiated connections as a function of the Layer 2 inhibitory synaptic time constant. The vertical axis represents the proportion of potentiated synapses relative to the total number of initial connections for both feedforward (FF, blue) and feedback (FB, red) projections. The number of stabilized FB connections decreases as the resonance frequency of the second layer network deviates more from that of the first layer network.
(B) Connection delays across varying inhibitory time constants in the layer 2 network. Weighted average of axonal delays for potentiated feedforward (FF, blue) and feedback (FB, red) connections. The total loop delay (dashed black) increases as the resonance frequency of the second layer decreases.

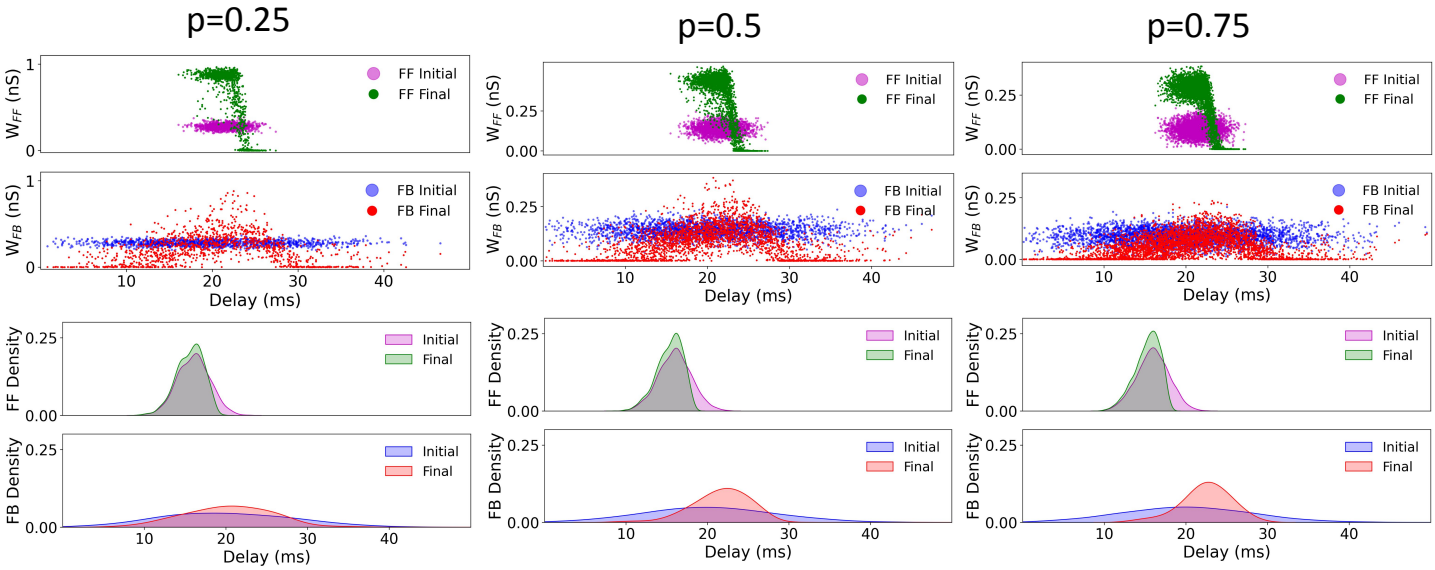


Figure S4: Delay selection scales with inter-layer connection probability. (Upper two rows) Evolution of synaptic weights and axonal delays under varying projection density. Scatter plots compare initial (magenta/blue) and final (green/red) states for feedforward (FF) and feedback (FB) connections, respectively. **(Lower two rows)** Delay distributions. The FF pathway exhibits a consistently sharp delay cutoff at 22–24 ms across all connection probabilities. However, the FB pathway demonstrates a moderate loss of structural precision as the connectivity becomes sparser. Synaptic weights and the maximum STDP conductance (W_{max}) were scaled inversely with connection probability to maintain a constant total input drive across different connectivity conditions.

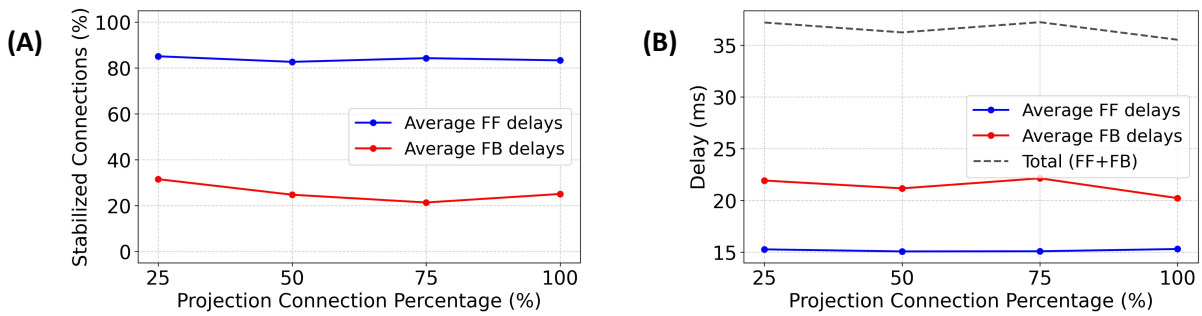


Figure S5: (A) Dependence of Synaptic Stabilization on Projection Density. The percentage of feedforward (FF, blue) and feedback (FB, red) connections that reached a stable potentiated state across varying initial projection densities (25% to 100%). To ensure a fair comparison across scales, adaptive stabilization thresholds were utilized for each condition (0.6 nS for 25% density, down to 0.15 nS for 100% density), effectively normalizing for total potential synaptic drive. **(B) Stability of delay selection across varying projection densities.** The weighted average of axonal delays for stabilized feedforward (FF, blue) and feedback (FB, red) connections as a function of the initial projection connection density. To account for varying network densities, adaptive weight thresholds were applied (0.6 nS for 25% density, down to 0.15 nS for 100% density). The total loop delay (dashed black curve), representing the sum of FF and FB averages, remains approximately constant, indicating that the network's structural resonance is robust to changes in projection density.